

Javier Sempere Hernández

Data Scientist & ML Engineer

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ML Engineer with a strong mathematical background (BSc Mathematical Engineering, MSc Computational Statistics) and 3+ years of industry experience building and deploying machine learning systems in production. Hands-on expertise in generative models, federated learning, and MLOps. Track record of leading technical projects as consultant and researcher, working directly with clients in healthcare, cybersecurity, and space. Experienced in privacy-sensitive applications and scalable AI within microservices architectures.

Spanish: Native | **English:** C1

WORK EXPERIENCE

ML Engineer — *GMV GmbH – Darmstadt, Germany* May 2025 – Present

- Led as sole consultant the end-to-end delivery of a Google Cloud-based ETL and BI pipeline for DLP analysis at a major European bank, now in production monitoring 5,000+ endpoints. Implemented an AI agent that autonomously generates periodic security-status reports for the bank's security officers, reducing manual reporting workload. Managed direct client engagement throughout, adapting the solution to evolving requirements.
- Led technical proposal writing for AI research and engineering projects submitted to the European Space Agency (ESA).
- Designed and implemented a VAE-based synthetic tabular data generator, enabling compliant data sharing under GDPR constraints.
- Built an LLM-powered AI agent integrated into a microservices architecture via Dapr, improving system modularity and enabling seamless inter-service communication.

Data Scientist — *GMV – Madrid, Spain* Jan 2024 – Apr 2025

- Main developer of a production-ready federated learning platform (Python, PyTorch, gRPC, European Data Space connectors) for distributed training across clinical nodes; supported clients in developing deep learning models for medical applications.
- Contributed to VALLE, an European Space Agency (ESA) NAVISP Element 1 project, implementing privacy-preserving PNT techniques via homomorphic encryption and SMPC in Zig and Python for encrypted GNSS signal processing.
- Led an anomaly-detection project for industrial cybersecurity environments in collaboration with a research team from Universidad de Castilla-La Mancha (UCLM).
- Developed and deployed discriminative and generative ML models in production.

Data Scientist Intern — *GMV – Madrid, Spain* Jun 2023 – Dec 2023

- Researched Privacy Enhancing Technologies: differential privacy, secure multi-party computation, and federated learning applied to sensitive data pipelines.
- Developed a Python-based tooling component for a European Data Spaces platform, contributing to cross-organisational data sharing infrastructure.

EDUCATION

MSc in Computational Statistics — *Universidad Politécnica de Madrid* 2023 – 2024

Convex optimisation, neural networks, Monte Carlo simulation, time series, and probabilistic graphical models. 60 ECTS.

BSc in Mathematical Engineering — *Universidad Complutense de Madrid* 2019 – 2023

Advanced pure and applied mathematics. Community of Madrid Excellence Scholar. 240 ECTS.

AWARDS & RECOGNITION

- Selected for the inaugural ESA Academy Cybersecurity Training Course (Apr 2024, ESA ESEC, Belgium).
- 1st place, Mathematical Modelling Competition UCM 2021–22 – automated defect detection in textile manufacturing using statistical inference (problem proposed by GMV; led to a job offer).
- Finalist, Spanish Programming Olympiad (2019) – Olympiad aimed at those under 18 years of age. Competitive programming problems solved in C++.